

ICT2203 Network Security



AY2021-2022 Trimester 1

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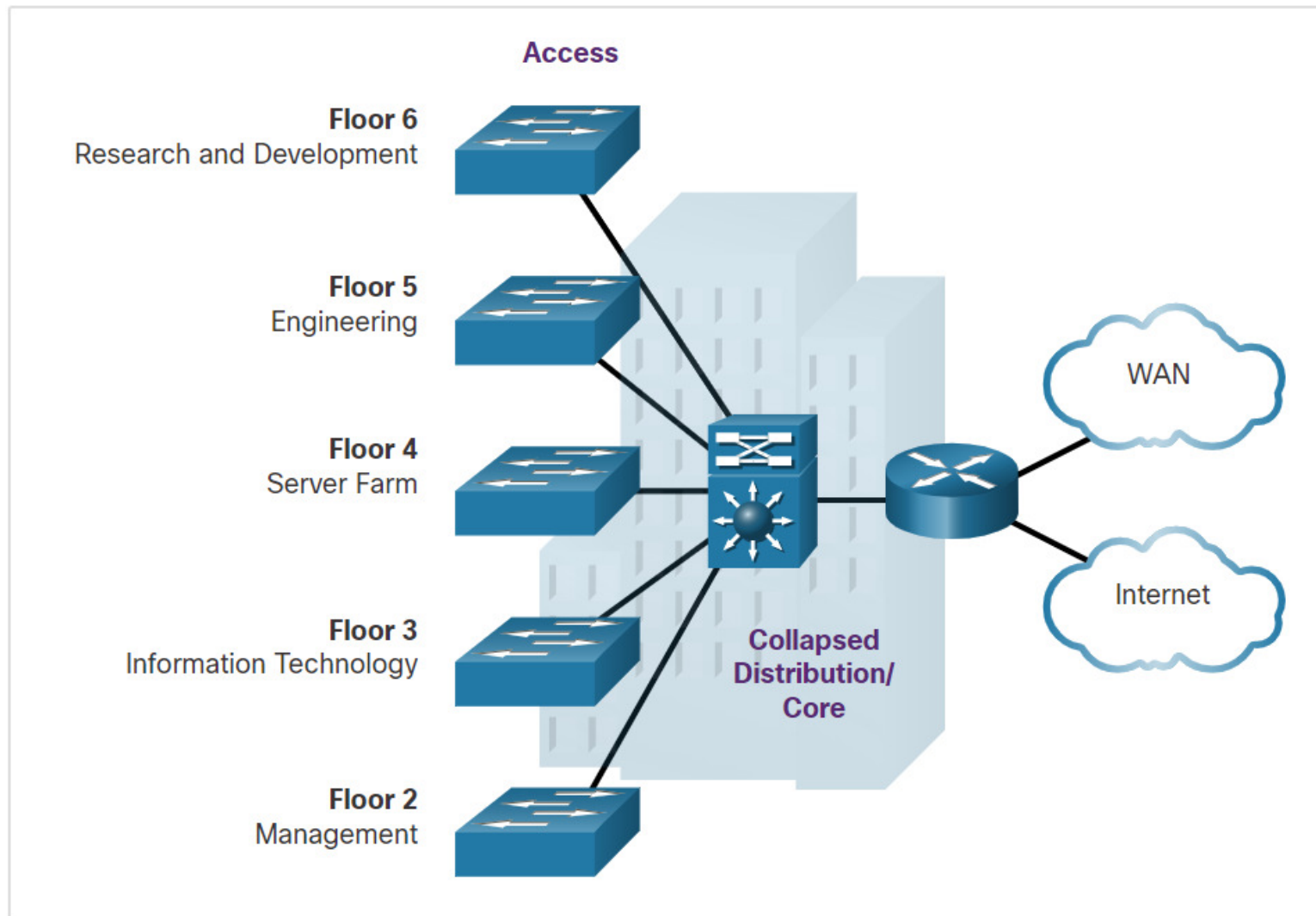
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Lab Professional Officers:

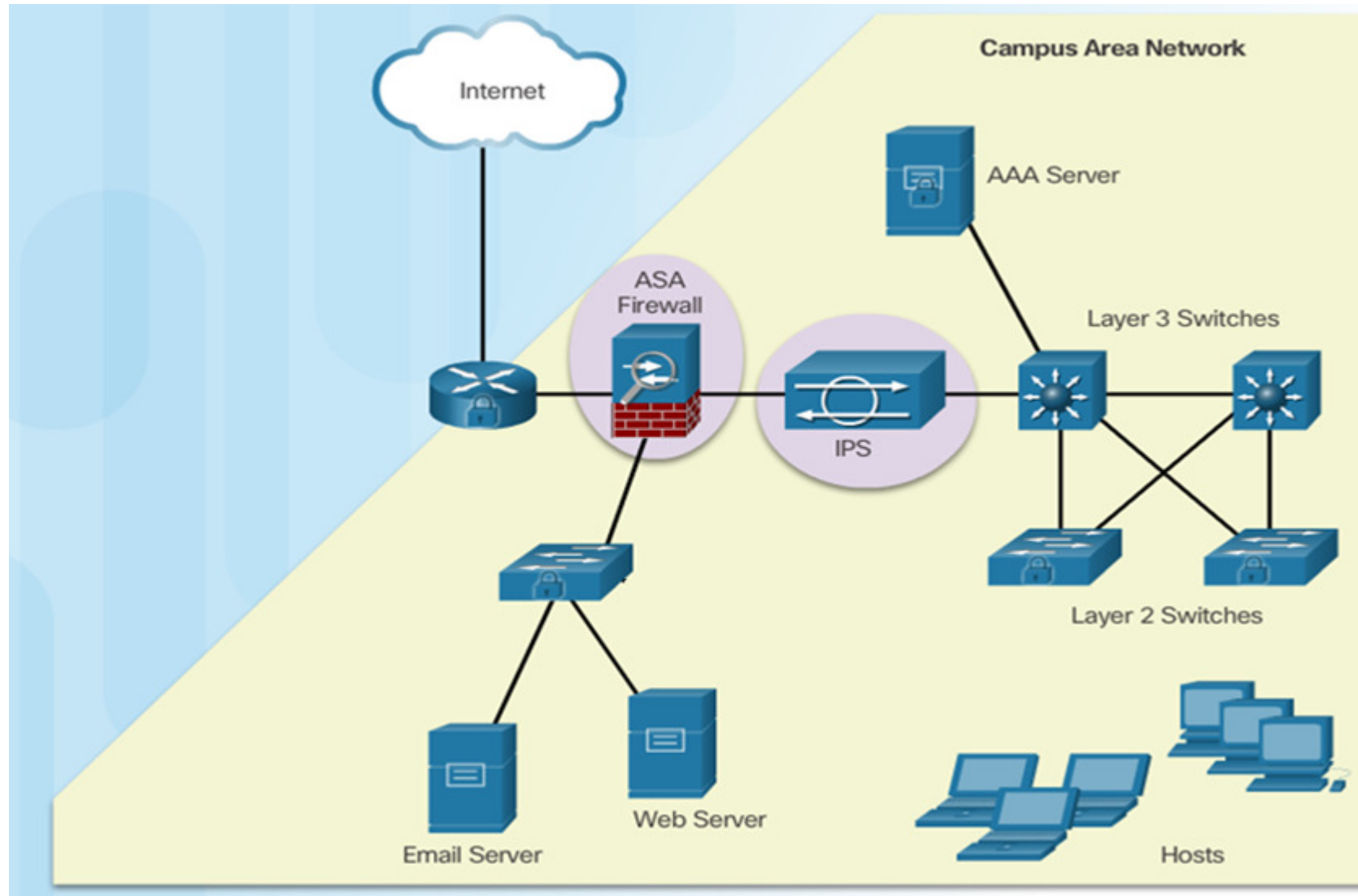
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In ICT1010, you have learned the design and implementation of enterprise network.



In ICT2203, you are going to learn the **attacks** and **defences** of the enterprise network.



Learning outcomes of ICT2203:

1. Explain and demonstrate common attacks on computer networks.

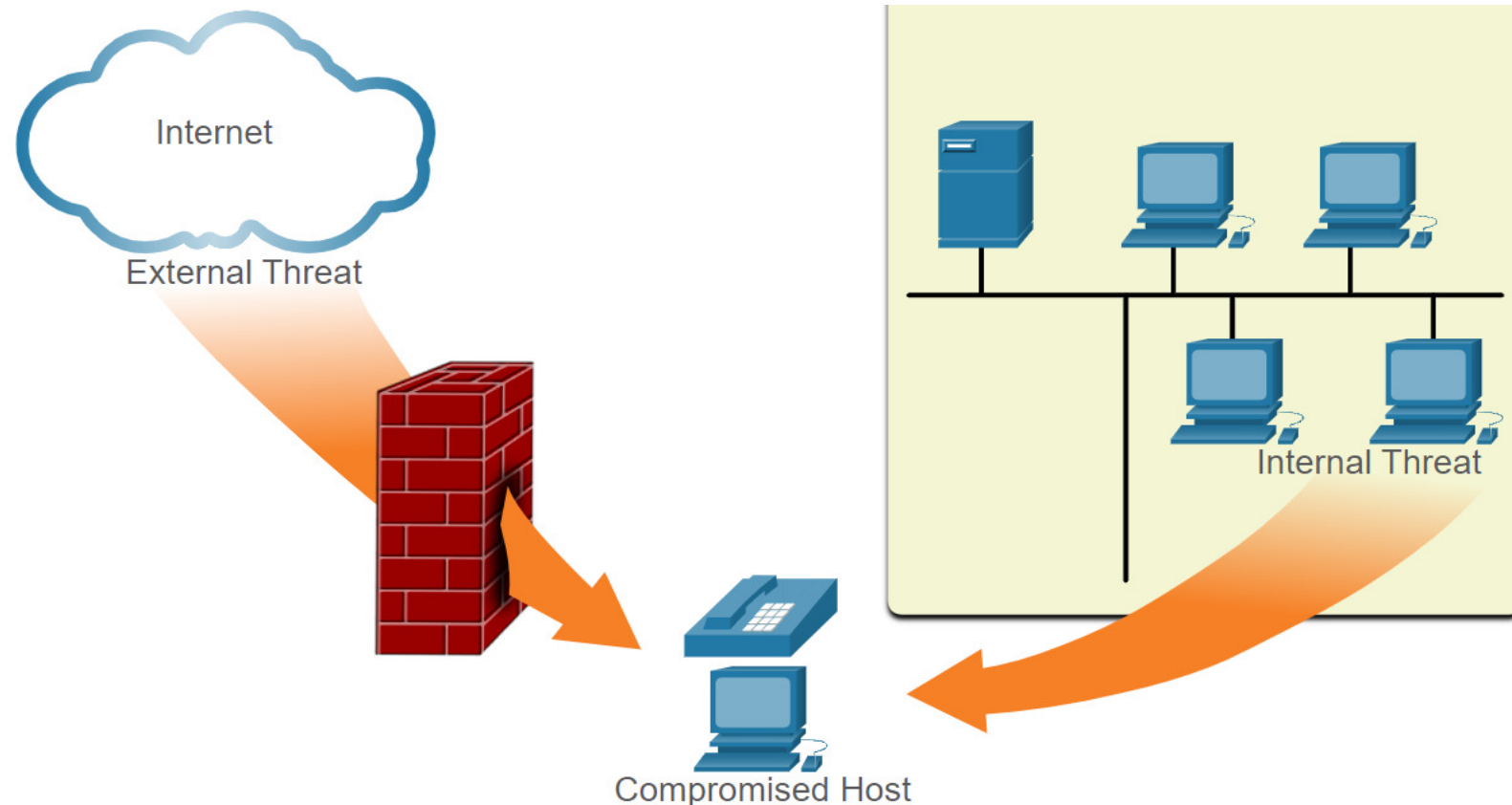
2. Implement defense-in-depth with switches, routers, firewalls and IDS/IPS.

3. Implement network device access control with AAA.

4. Perform network security monitoring with NTP, Syslog, SNMP and NetFlow.

In practice, **threat actor** (**hacker** or **nation state**) may attack from **inside** or **outside** the enterprise network.

- **External**: from unauthorized individuals who are trying to gain access to the network, typically from the Internet
- **Internal**: from authorized users who have access to the enterprise network, which can be **common** and **difficult to defend**



Hence, the topics of ICT2203 are organized from internal to external attacks and defenses as shown.

Labs
• Attacks and Defense of LAN with Switches
• More Attacks and Defense of LAN with Switches
• Attacks and Defense of IP Networks with Routers
• Network Segmentation and Access Control with Firewalls
• Intrusion Detection and Prevention with IDS/IPS
• Device Access Control with AAA
• Network Security Monitoring with NTP, Syslog, NetFlow and SNMP

ICT2203 will be conducted fully in labs using real devices for you to acquire knowledge and skills
= in **attacking** and **defending** enterprise network.



Important security ethics and **warning**: Attacking real world production networks without owner's explicit written permission can get you into **real troubles**!

Time-table for Part I of ICT1003:

Sun	Mon	Tue	Wed	Thu	Fri	Sat
5 Sep Week 1	6 Grp P1/P2: Lec/Lab 1 13:30 - 18:30	7 Grp P3/P4: Lec/Lab 1 13:00 - 18:00	8	9	10	11
12 Sep Week 2	13 Grp P1/P2: Lec/Lab 2 13:30 - 18:30	14 Grp P3/P4: Lec/Lab 2 13:00 - 18:00	15	16	17	18
19 Sep Week 3	20 Grp P1/P2: Lec/Lab 3 13:30 - 18:30	21 Grp P3/P4: Lec/Lab 3 13:00 - 18:00	22	23	24	25
26 Sep Week 4	27 Grp P1/P2: Lec/Lab 4 13:30 - 18:30	28 Grp P3/P4: Lec/Lab 4 13:00 - 18:00	29	30	1 Oct	2
3 Oct Week 5	4 Grp P1/P2: Lec/Lab 5 13:30 - 18:30	5 Grp P3/P4: Lec/Lab 5 13:00 - 18:00	6	7	8	9
10 Oct Week 6	11 (Part II) Lec (Grp P1/P2/P3/P4) 12:00 - 14:00 Grp P1/P2: Lab 6 15:30 - 18:30	12 (Part II) Grp P3/P4: Lab 6 13:00 - 16:00 Quiz 1 (Grp P1/P2/P3/P4) 17:00 - 18:15	13	14	15	16
17 Oct Week 7	18 Recess	19 Recess	20 Recess	21 Recess	22 Recess	23
24 Oct Week 8	25 (Part I) Grp P1/P2: Lec/Lab 7 13:30 - 18:30 Assignment 1 Demo	26 (Part I) Grp P3/P4: Lec/Lab 7 13:00 - 18:00 Assignment 1 Demo	27	28	29	30

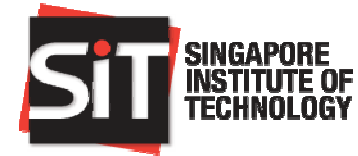
Assessments of ICT2203:



**ASSESSMENT
INFO**

	Assessments	Weighting
1a.	Assignment	20%
1b.	Peer Appraisal 1 for Assignment	Multiplier factor to derive individual grade
2.	Quizzes 1 & 2	30%
3.	Practical Test	20%
4a.	Team Project	30%
4b.	Peer Appraisal 2 for Team Project	Multiplier factor to derive individual grade
	No Final Exam	0%
	Total	100%

All materials for **ICT2203** will be posted at
<https://xsite.singaporetech.edu.sg>



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ICT2203-Network Security [2021/22 T1]

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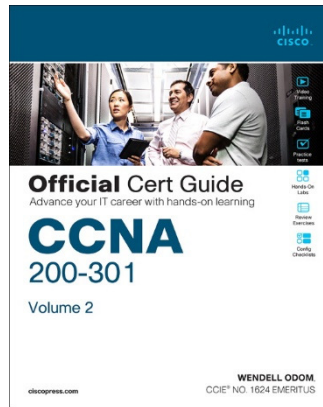
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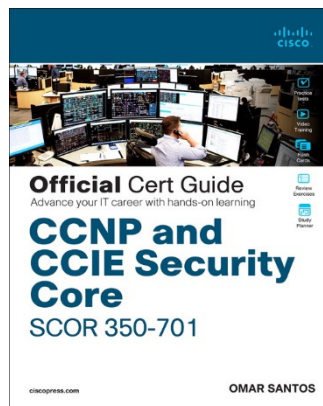
📁 [Lab 0: Getting to know Kali Linux](#) >

📁 [Lab 1: Attacks and Defense of Local Area Networks \(LAN\) with Switches Part 1](#) >

Besides lab materials, you may wish to refer other [references](#), e.g.:



CCNA 200-301 Official Cert Guide, Volume 2
by Wendell Odom
Cisco Press, 2020



CCNP and CCIE Security Core SCOR 350-701
Official Cert Guide
by Omar Santos
Cisco Press, 2020